

ROSEMARY AGDM, AB, Canada

WMO: 715370

Lat: 50.8344N

Lon: 112.0573W

Elev: 2392

StdP: 13.47

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-26.7	-19.8	-32.6	1.0	-26.6	-25.4	1.5	-19.6	26.8	29.1	24.3	32.0	5.3	250	0.688

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.8	87.5	63.9	84.0	62.6	80.8	61.6	67.1	81.9	65.2	79.5	63.4	77.1	8.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.9	90.4	74.1	59.7	83.6	71.4	57.8	78.0	69.1	33.0	81.2	31.4	79.5	30.0	77.5	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.5	21.8	18.8	DB	-36.1	93.8	7.1	2.9	-41.3	95.9	-45.4	97.6	-49.5	99.2	-54.6	101.3	(4)
(5)			WB	-36.4	72.1	7.4	1.5	-41.7	73.2	-46.0	74.1	-50.2	74.9	-55.5	76.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.7	14.3	13.6	26.9	40.1	51.6	58.9	64.6	61.8	52.9	40.0	25.6	12.5		
	DBStd	21.70	15.68	15.19	14.42	8.96	7.39	5.30	4.99	5.35	7.59	8.98	13.52	15.32		
	HDD50	5502	1106	1019	717	308	69	3	0	1	54	328	734	1163		
	HDD65	9696	1571	1439	1181	747	415	196	69	126	366	776	1182	1628		
	CDD50	1380	0	0	1	10	120	269	452	368	141	17	2	0		
	CDD65	100	0	0	0	0	1	12	55	28	4	0	0	0		
	CDH74	2908	0	0	0	16	200	346	1180	845	309	12	0	0		
	CDH80	854	0	0	0	2	40	79	385	256	92	0	0	0		
(14) Wind	WSAvg	8.3	8.1	7.5	8.7	9.7	9.7	8.8	7.8	7.5	7.9	8.4	8.3	7.5		
(15) Precipitation	PrecAvg	13.8	0.5	0.4	0.7	1.2	2.1	2.8	1.7	1.5	1.1	0.7	0.6	0.5		
	PrecMax	21.0	1.0	0.9	2.2	2.7	3.6	7.9	4.0	3.7	4.3	1.9	1.5	1.1		
	PrecMin	6.3	0.0	0.0	0.0	0.3	0.5	0.9	0.3	0.0	0.1	0.1	0.0	0.0		
	PrecStd	2.6	0.2	0.2	0.4	0.6	0.9	1.5	1.0	0.9	0.9	0.4	0.4	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.2	50.8	64.0	75.8	84.7	87.5	92.5	90.6	88.5	75.5	64.1	49.9		
		MCWB	40.0	40.6	47.5	52.0	58.3	65.2	66.3	63.3	61.0	53.9	49.0	40.3		
	2%	DB	43.8	44.3	58.2	69.3	79.2	81.8	88.1	86.3	82.1	68.2	55.3	43.5		
		MCWB	37.1	37.1	44.3	49.5	56.0	61.4	66.4	63.0	59.2	51.4	43.7	37.1		
	5%	DB	39.7	39.1	52.7	63.7	74.9	77.8	84.2	82.5	76.8	63.4	49.4	39.2		
		MCWB	35.0	34.3	41.6	46.6	54.9	59.9	64.6	62.6	57.2	49.1	40.0	34.7		
	10%	DB	35.9	34.4	46.3	58.9	70.2	74.2	80.7	78.8	71.8	57.7	44.4	34.7		
		MCWB	32.4	31.4	38.3	44.2	52.1	58.0	63.2	61.1	55.1	46.0	37.1	31.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.1	41.1	48.7	53.4	60.0	66.9	70.8	68.5	63.3	56.0	48.9	41.4		
		MCDB	48.0	48.3	62.2	73.1	78.7	81.3	85.1	82.7	82.1	71.5	63.3	47.9		
	2%	WB	37.7	36.7	45.3	49.9	57.5	64.2	68.1	65.8	60.2	52.4	44.3	37.4		
		MCDB	42.8	42.1	56.5	67.7	76.1	78.3	83.0	80.5	78.3	66.0	54.5	42.5		
	5%	WB	35.3	34.3	41.7	47.4	55.7	61.6	66.3	63.8	58.1	49.5	41.0	34.5		
		MCDB	39.8	38.3	51.5	61.5	72.0	74.0	81.1	78.5	74.9	62.1	48.5	39.0		
	10%	WB	32.7	30.9	38.4	44.8	53.5	59.6	64.5	62.1	55.7	46.8	37.5	31.1		
		MCDB	36.2	33.9	45.4	56.7	67.7	71.2	77.8	76.1	70.5	56.8	43.1	34.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.8	23.6	23.9	26.9	28.4	25.5	28.8	29.5	30.2	27.7	23.5	21.8		
		MCDBR	24.0	27.8	32.3	39.0	37.9	33.0	36.0	38.6	41.8	36.9	30.4	25.3		
		MCWBR	18.2	20.6	19.1	20.4	17.2	15.7	16.6	18.0	19.9	20.7	19.8	19.1		
	5% WB	MCDBR	23.0	26.1	30.2	36.3	34.1	30.0	32.6	34.0	39.1	35.1	29.4	25.4		
		MCWBR	18.0	20.4	18.9	19.5	16.1	15.6	16.3	17.0	19.1	20.4	19.9	19.6		
(40) Clear-Sky Solar Irradiance	taub		0.271	0.285	0.298	0.350	0.376	0.366	0.364	0.383	0.334	0.293	0.264	0.254		
	taud		2.375	2.414	2.444	2.302	2.278	2.357	2.392	2.310	2.428	2.520	2.485	2.395		
	Ebn at Noon		244	273	289	282	279	282	281	268	272	267	249	235		
	Edh at Noon		19	24	29	37	40	37	36	36	29	22	17	16		
(44) All-Sky Solar Radiation	RadAvg		374	692	1122	1538	1841	1977	2138	1724	1228	750	407	285		
	RadStd		33	62	84	113	125	121	102	115	113	85	36	26		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (82 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page